Telecom Data Analysis

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Project Overview

- **Project Objective:**

Analyze customer attrition patterns to identify root causes of churn.

Evaluate network performance impact on customer satisfaction (CSAT).

Provide data-driven recommendations to improve retention and revenue.

- **Key Metrics Tracked:**

Churn Rate %.

ARPU (Average Revenue Per User).

Drop Call Rate %.

Average Data Throughput

Network Availability %

Avg Resolution Time

CSAT (Customer Satisfaction Score)

High Value Customer Risk %

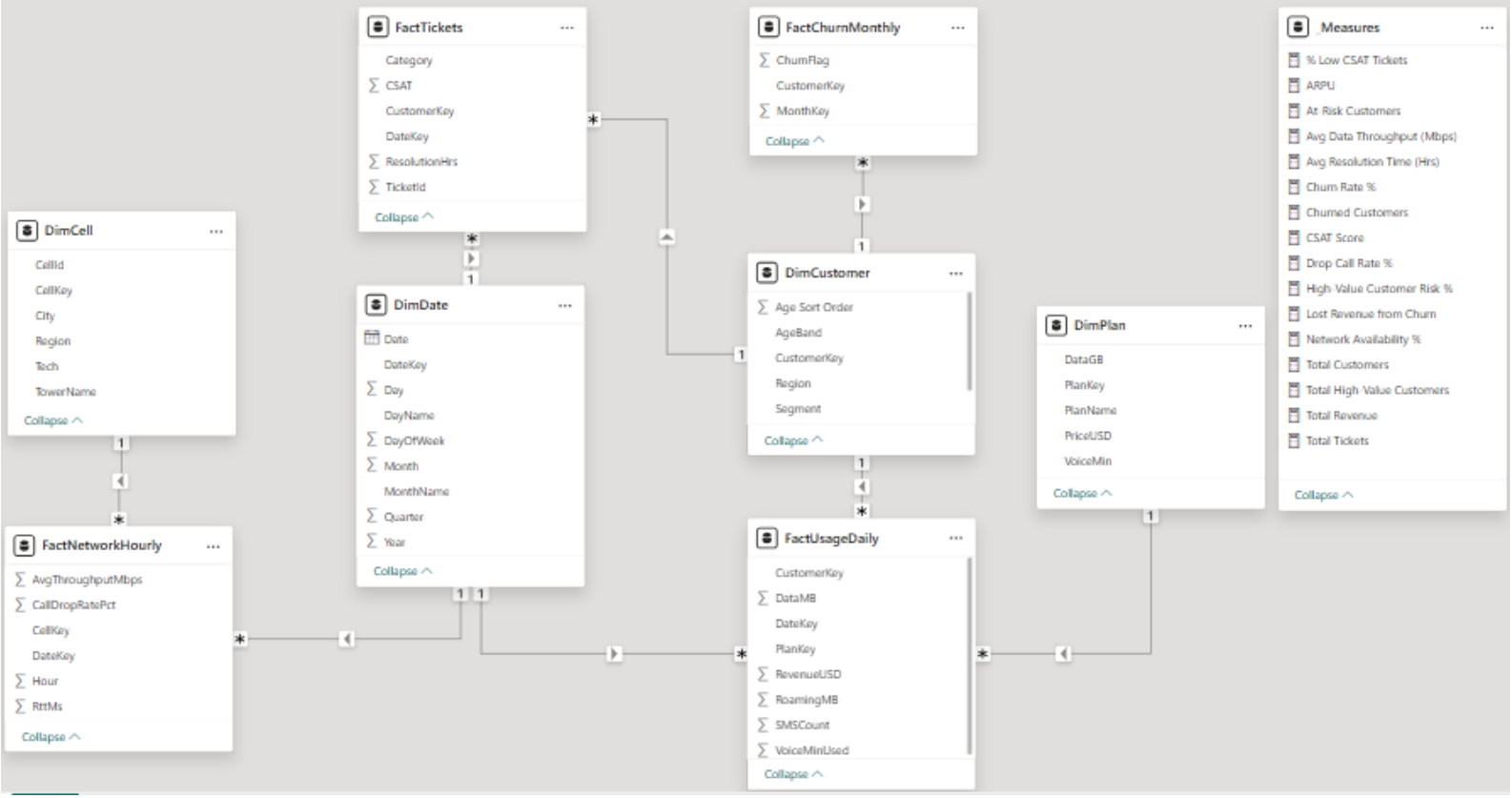
Total Revenue
Churned Customers
Total Customer
Lost Revenue from Churn
Total Tickets

Data Model & Methodology

- **Star Schema Design:** Centralized Fact tables (Usage, Churn, Tickets, Network) linked to Dimension tables (Customer, Date, Location).
- **ETL Process:** Utilized Power Query for data cleaning, transformation, and error handling.

Technical Challenge & Solution:

- **Challenge:** Missing link between FactUsage and DimPlan (No direct PlanKey).
- **Solution (SCD Type 2 Logic):** Implemented a **30-day Moving Average** of data usage in Power Query to infer the customer's plan historically.
    - *Result:* Accurate historical tracking of plan changes vs. usage, ensuring precise revenue analysis without data duplication.



Dashboard Highlights

- **Executive Summary:** High-level KPIs (Revenue: $52K, Churn: 44.33%) for immediate health check.
- **Churn Analysis:** Deep dive into churn by segment, tenure, and age band.
- **Network Operations:** Geospatial analysis of Drop Call Rates and Signal Strength.
- **Customer Support:** Correlation between Ticket Resolution Time and CSAT scores.
- **User Experience:**
- Dynamic filtering by Region, Month, and Customer Segment.
- Drill-down capabilities to identify specific risk groups.

Insights & Findings

- **Critical Discoveries:**
- **Alarming Churn Rate:** The churn rate is **44.33%**, significantly impacting total revenue ($23.72K lost revenue).
- **Network Issues in "West" Region:**
  - Highest **Drop Call Rate** observed in West Region.
  - Strong correlation with low CSAT scores in this specific area.
- **High-Value Customer Risk:**
  - "HighValue" and "SME" segments show concerning churn trends despite high spending.
  - Customers with **CSAT scores < 2** are leaving rapidly.
- **Support Efficiency:** Longer resolution times are directly driving CSAT down, accelerating churn.

Recommendations

- **Strategic Actions:**
- **Immediate Infrastructure Repair:** Dispatch network engineering teams to the **West Region** to fix coverage gaps and reduce Drop Call Rate.
- **Proactive Retention Program:**
  - Implement an automated alert system for **High-Value Customers** when CSAT drops below 3.
  - Offer targeted retention offers before they churn (churn prediction).
- **Support Process Optimization:**
  - Investigate "Billing" and "Tech" ticket categories which have the longest resolution times.
  - Set stricter SLAs for ticket closure to improve CSAT.
- **Next Steps:**
- Monitor the impact of West Region fixes on next month's Churn Rate.